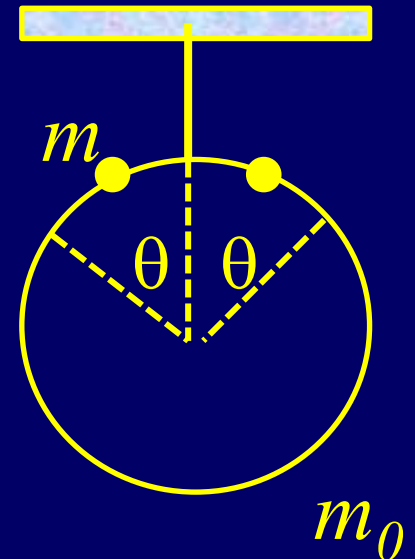


Quiz 9:

A large ring with mass m_0 is hung from the ceiling by a thin wire. Two small rings with m set in the large ring can move freely. If two small rings simultaneously release from the top of the large ring at rest, they will slip down oppositely. When two small rings slip to the angle θ , the large ring will start to rise up. Please calculate this angle and the minimum mass of the small rings required for rising the large ring.



Answer:

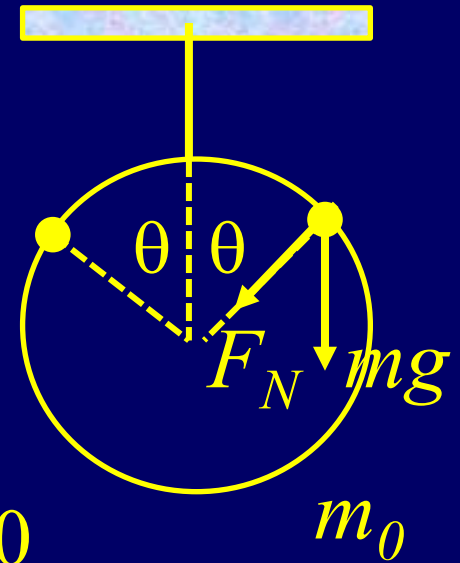
$$mgR(1 - \cos \theta) = \frac{1}{2}mv^2$$

$$mg \cos \theta + F_N = mv^2 / R$$

$$2F_N \cos \theta = m_0 g$$

$$6m \cos^2 \theta - 4m \cos \theta + m_0 = 0$$

$$\cos \theta = (1 \pm \sqrt{1 - 3m_0 / 2m}) / 3$$



We should choose the larger value which gives a smaller angle.

$$\cos \theta = (1 + \sqrt{1 - 3m_0 / 2m}) / 3$$

The condition $m_0 / m \leq 2 / 3$ should be satisfied.